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12 MARCH 1963

SUPERSEDING
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19 JULY 1962

MILITARY SPECIFICATION

CASEIN HYDROLYSATE

This specification has been approved by the Department of Defense and is mandatory for use by the Departments of the Army, the Navy, and the Air Force.

1. SCOPE

1.1 Scope. This specification covers two grades of casein hydrolysate.

1.2 Classification. The casein hydrolysate shall be of the following grades, as specified (see 6.2):

Grade A — Pancreatic hydrolysate (AN/TN 50%) (see 6.7).

Grade B — Pancreatic hydrolysate (AN/TN 40%) (see 6.7).

2. APPLICABLE DOCUMENTS

2.1 Government documents. The following document of the issue in effect on date of invitation for bids or request for proposal forms a part of this specification to the extent specified herein:

STANDARDS

MILITARY

MIL-STD-129 — Marking for Shipment and Storage.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

ASSOCIATION OF OFFICIAL AGRICULTURAL CHEMISTS

Official Methods of Analysis of the Association of Agricultural Chemists (8th edition).

(Copies of the AOAC methods of analysis may be obtained from the Association of Official Agricultural Chemists, P.O. Box 540, Benjamin Franklin Station, Washington 4, D.C.)

UNITED STATES PHARMACOPEIAL CONVENTION, INC.

The Pharmacopeia of the United States of America (U.S.P. XVI, 1960).

(Copies of the U.S.P. may be obtained from the Mack Publishing Company, Easton, Pa.)

AMERICAN SOCIETY FOR TESTING MATERIALS

ASTM Standards —

D1013 — Method of Test for Total Nitrogen in Resins for Surface Coatings.

E70 — Method for the Determination of the pH of Aqueous Solutions with the Glass Electrode.

(Application for copies should be addressed to the American Society for Testing Materials, 1916 Race St., Philadelphia 3, Pa.)

FSC 6810

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UNIFORM CLASSIFICATION COMMITTEE

Uniform Freight Classification Rules.

(Application for copies of these freight classification rules should be addressed to Uniform Classification Committee, 202 Union Station, Chicago 6, Ill.)

3. REQUIREMENTS

3.1 Form. The casein hydrolysate shall be in the form of a dry powder.

3.2 Amino acid assay (proximate). The proximate assay of the amino acids in the casein hydrolysate shall conform to the requirements specified in table I.

3.3 Nutritional value. The casein hydrolysate shall supply sufficient nutritional value to support adequate growth of various microorganisms (see 6.3) when tested as specified in 4.5.1.

3.4 Pre-award sample. At no cost to the Government, each bidder shall forward a sample of the lot of casein hydrolysate intended for shipment under the contract or order to the destination indicated by the contracting officer. The Government will test the samples to determine compliance with the requirements of this specification.

TABLE I. *Amino acid assay (proximate)*

Acid	Requirement	
	Grade A	Grade B
Glycine	1.6%	1.8%
Alanine	3.1%	3.1%
Valine	6.0%	6.1%
Leucine	8.6%	9.4%
Isoleucine	4.1%	4.6%
Serine	5.7%	6.2%
Threonine	3.7%	4.2%
Lycine	7.4%	7.6%
Arginine	3.8%	3.9%
Aspartic	5.5%	5.9%
Glutamic	19.7%	19.6%
Proline	11.0%	10.8%
Histidine	2.8%	2.9%
Tryptophane	1.4%	1.6%
Phenylalanine	5.1%	5.0%
Tryosine	3.1%	3.2%
Cystine	0.42%	0.46%
Methionine	2.3%	2.4%

3.5 Chemical and physical properties. The chemical and physical properties of the casein hydrolysate shall conform to the requirements specified in table II when tested in accordance with the test paragraphs referenced therein.

TABLE II. *Chemical and physical properties*

Property	Requirement				Test paragraph
	Grade A		Grade B		
	min	max	min	max	
Moisture (%)	3.0	6.0	3.0	6.0	4.5.2
Ash (%)		5.9		5.8	4.5.3
Total nitrogen (%)	12.0	16.0	12.0	16.0	4.5.4
Amino nitrogen (%)	6.4	6.9	5.1	5.6	4.5.5
Amino N/Total N (%)	50.0	53.0	40.0	43.0	4.5.6
Sodium (%)	1.6	1.8	1.6	1.8	4.5.7
Chloride (as NaCl) (%)		0.3		0.3	4.5.8
Iron (ppm)		50		50	4.5.9
Heavy metals (as Pb) (ppm)		20		20	4.5.10
Copper (ppm)		7		7	4.5.11
ph (2% solution at 25° C.)	6.6	6.9	6.9	7.2	4.5.12
Solubility (at 30° C.)	pass test		pass test		4.5.13
Bacteria (per g)		10,000		10,000	4.5.14

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3.6 Acceptance lots. The lot or lots submitted to the Government for final acceptance shall be the lot or lots from which the acceptance samples (see 4.4.2) were selected and approved by the Government as complying with the requirements of this specification.

3.7 Storage conditions. During the period between the taking of the acceptance samples and the time of shipment, the lot or lots shall not be subjected to any known condition which may have adverse effects on the nutritional value of the casein hydrolysate.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.

4.1.1 *Supplier's responsibility.* Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.2 *Government responsibility.* The Government will perform the tests specified in 4.5.1.

4.2 Special provisions.

4.2.1 *Alternative inspection (including testing) procedures.* The supplier may utilize any alternative inspection procedures which will insure equal or better quality by submitting a written proposal with justification and obtaining written approval from the Government prior to instituting the procedure. In case of dispute, the procedures of this specification will govern.

4.2.2 *Objective evidence.* The supplier shall provide evidence acceptable to the contracting officer that the requirements of 3.1, 3.2, 3.4, 3.6, 3.7 and section 5 have been satisfied.

4.3 Lotting. A lot shall consist of the casein hydrolysate offered for acceptance at one time, which has been produced by one manufacturer under essentially the same manufacturing conditions and with no change in materials provided the operation is continuous. In the event that the process is a batch operation, each batch shall constitute a lot (see 6.4).

4.4 Sampling.

4.4.1 *Pre-award sample.* A 10-pound representative sample of the casein hydrolysate shall be placed in a clean, dry container labeled in accordance with Standard MIL–STD–129.

4.4.2 *Acceptance sample.* Sample containers shall be taken at random from each lot as follows:

Lot size	*Sample size*
1	1
2– 275	2
276– 545	3
546- 900	4
901–1345	5
1346–1875	6
1876–2500	7

Sample size for lots exceeding 2500 containers shall be calculated using the following equation:

$$n = 0.15\sqrt{N}$$

where: n = sample size.
N = lot size.

A 1-pound specimen shall be removed from each container in the sample and placed in a clean, dry container labeled to identify the lot and the container from which it was taken. Each specimen shall be tested as specified in 4.5.

4.5 Tests. Distilled water and analytical reagent or biological grade chemicals shall be used throughout the tests. Where applicable, blank determinations shall be run and correc-

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tions applied when significant. Tests shall be conducted as follows:

4.5.1 *Nutritional value.* Combine the casein hydrolysate with known standard materials to form a medium and determine its nutritional value by means of the growth rate of various micro-organisms.

4.5.2 *Moisture.* Adjust a direct reading, infra-red heated, moisture balance to 100 percent with the weighing dish in place. Turn the dial to zero and add a sufficient quantity of the specimen (see 4.4.2) to restore exact balance. (This assures a sample weight of 10.0 grams (g).) Close the hood containing the heating unit and apply about 125 watts to the heater. Continue the heating until the pointer fails to show a loss in weight greater than 0.1 percent in a period of 3 minutes. Read the percent moisture directly from the scale.

4.5.3 *Ash.* Weigh 2.0 g of the specimen to an accuracy of 1 milligram (mg) in a previously tared platinum crucible. Place the crucible and contents in an open-coil electric heater, or over a gas burner, and char the contents until smoke no longer comes off. Place the crucible in a muffle furnace at about 650° C. for not less than 5 hours. At the end of the heating period, remove the crucible and place it in a desiccator. After the crucible has cooled, weigh it to an accuracy of 1 mg (save the ash for the heavy metals test specified in 4.5.10). Calculate the percent ash using the formula:

$$\text{Percent ash} = \frac{100A}{W}$$

Where: A = weight of ash.
W = weight of specimen used.

4.5.4 *Total nitrogen.* Determine the percent total nitrogen by means of the test specified in ASTM Designation D1013.

4.5.5 *Amino nitrogen.* Determine the percent amino nitrogen by means of the test specified for *a*-amino nitrogen in Protein Hydrolysate Injection, USP, XVI, page 601, except that the USP formaldehyde T.S. shall have been previously adjusted to pH 7.0.

4.5.6 *Amino nitrogen/total nitrogen.* Determine the percent amino nitrogen in the total nitrogen by dividing the percent amino nitrogen (see 4.5.5) by the percent total nitrogen (see 4.5.4) and multiplying the quotient by 100.

4.5.7 *Sodium.* Determine the percent sodium by means of a flame photometer, following the instructions which accompany the photometer.

4.5.8 *Chloride (as NaCl).*

4.5.8.1 *Solutions.*

4.5.8.1.1 *Sodium chloride solution (0.1N).* Place about 10 g of sodium chloride in a platinum vessel, heat it to dull redness for approximately 15 minutes, and, after cooling, grind the calcined salt to a fine powder. Weigh 5.845 g of the powder, transfer it to a 1-liter flask, fill the flask to the mark with water, and thoroughly mix the solution.

4.5.8.1.2 *Silver nitrate solution (0.1N).* Weigh 17.1 g of silver nitrate and transfer it to a 1-liter volumetric flask. Fill the flask to the mark with water and thoroughly mix the solution. Determine the exact normality of the solution by titration against the 0.1*N* sodium chloride solution. Store the solution in a glass-stoppered brown bottle. Protect the bottle from direct sunlight.

4.5.8.1.3 *Potassium chromate solution.* Prepare a saturated potassium chromate solution.

4.5.8.2 *Procedure.* Weigh from 1.5 to 2.0 g of the specimen to an accuracy of 1 mg and transfer it to a 250 milliliter (ml) volumetric flask. Fill the flask to the mark with water and mix until solution is effected. Using a pipette, transfer 50-ml aliquots to separate clean porcelain casseroles. Add 2 or 3 drops of the potassium chromate solution and titrate each aliquot with the silver nitrate solution. Perform a standardization titration in parallel with the specimen titrations to

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the same depth of red coloration, and correct the normality of the silver nitrate solution. Calculate the percent chloride as NaCl using the formula:

$$\text{Percent chloride (as NaCl)} = \frac{5.845\,AN}{W}$$

where: A = average ml 0.1*N* silver nitrate required for aliquots.
N = normality of silver nitrate solution.
W = weight of specimen in aliquots (⅕ of weight of specimen used).

4.5.9 *Iron.* Determine the parts per million (ppm) of iron by means of the test procedures specified in paragraphs 13.11, 13.12, and 13.13 of Official Methods of Analysis of Official Agricultural Chemists, Eighth Edition, or by means of a flame photometer using the instructions which accompany the photometer.

4.5.10 *Heavy metals (as Pb).* Add approximately 2 ml of 3*N* (approximately 10 percent) hydrochloric acid to the contents of the crucible used for the determination of ash (see 4.5.3) and digest the contents until solution is effected. Add dilute ammonium hydroxide until the solution is alkaline. Add 1*N* acetic acid until the solution is acid and then add 2 ml excess. Rinse the contents of the crucible into a Nessler tube, filtering if necessary, using small amounts of wash water. Determine the ppm of heavy metals by means of the test procedure specified in method II of U.S.P. XVI, p. 899.

4.5.11 *Copper* (see *6.6*). Transfer 5 g of the specimen weighted to an accuracy of 1 mg into a porcelain crucible, and carefully burn off the combustible matter over an open-coil electric heater or a gas burner. Place the crucible in a muffle furnace and ignite at about 650° C. for not less than 5 hours. At the end of the ignition period remove the crucible and allow it to cool. Add about 10 ml of water and 5 ml of concentrated hydrochloric acid. Heat the crucible on a steam bath for 10 to 15 minutes, and then filter the contents into a 100-ml volumetric flask, washing the residue and filter paper thoroughly with hot water. Cool the filtrate, fill the flask to the mark with water and thoroughly mix the solution. Pipette an aliquot of the solution, containing 0.015 to 0.030 mg of copper into a 125-ml separatory funnel and add 5 ml of a 15-percent citric acid solution. Add one drop of a 0.1-percent thymol blue solution and sufficient dilute ammonium hydroxide to obtain a faint blue color. Add about 10 ml of a 0.1-percent solution of sodium diethyldithiocarbamate in water and 10 ml of carbon tetrachloride and vigorously shake the funnel for 2 to 3 minutes. Allow the liquids to separate and draw off the carbon tetrachloride into a dry 25-ml erlenmeyer flask. Repeat the extraction twice, or until the carbon tetrachloride remains colorless, using 3 to 4 ml portions of carbon tetrachloride for each extraction and combining the extracts. Filter the combined extracts through a small quantity of anhydrous sodium sulfate into a 25-ml volumetric flask. Wash the filter with carbon tetrachloride, fill the flask to the mark with carbon tetrachloride, and thoroughly mix the solution. Determine the absorbence at 435 millimicrons by means of a suitable spectrophotometer using carbon tetrachloride as a blank. Using volumetric techniques, carry standard copper solutions through the above procedure, and plot the absorbence against the ppm of copper. (A convenient starting point may be prepared by dissolving 0.7860 gm of copper sulfate ($CuSO_4 \cdot 5H_2O$) in sufficient water to make 1,000 ml, and diluting 25 ml of this solution to 500 ml. The final solution will contain 0.01 milligrams of copper per ml.) Calculate the ppm copper using the equation:

$$\text{ppm copper} = \frac{2500\,M}{VW}$$

where: M = ppm copper from curve.
V = volume of aliquot used.
W = weight of specimen used.

4.5.12 *pH.* Dissolve 2.00 ± 0.01 g of the specimen in sufficient water to make 100 ml, and adjust the temperature to 25° ± 1°C.

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Determine the pH of the solution using the procedure specified in ASTM Designation: E70.

4.5.13 *Solubility.* Weigh 35.0 ± 0.1 g of the specimen and transfer it to a stoppered bottle containing 1,000 ml of water. Place the bottle on a shaking machine for 10 minutes maintaining a solution temperature of 30° ± 1° C. Examine the solution visually against a white background. Report the specimen as complying with the requirements if no cloudiness or undissolved particles are observed.

4.5.14 *Bacteria.*

4.5.14.1 *Milk protein hydrolysate agar.* Thoroughly mix the ingredients in the percentages specified in table III with the calculated quantity of water. Adjust the pH to 7.0 ± 1 by adding 2*N* sulfuric acid or 2*N* sodium hydroxide as required. Autoclave the suspension at 121° ± 1° C. for 20 minutes.

TABLE III. *Milk protein hydrolysate agar (see 6.5)*

Ingredient	Percent by weight
Milk protein hydrolysate	0.9
Dextrose	0.1
Agar	1.5

4.5.14.2 *Procedure.* Weigh 2.00 ± 0.01 g of the specimen and transfer it to a sterile bottle containing 20.0 ml of sterile water and a few sterile glass beads. Using aseptic procedures, cap the bottle and shake it until the specimen is dispersed. Using a sterile pipette, transfer 0.1 ml of the dispersion into each of two sterile 100 by 15 millimeter petri dishes, and 1.0 ml into a dilution bottle containing 99 ml of sterile water. Shake the dilution bottle thoroughly and, using a sterile pipette, transfer 1.0 ml of the dilution into each of two 100 by 15 millimeter sterile petri dishes. (Each of the first pair of dishes contains 0.01 g of the specimen and each of the second pair contains 0.001 g.) Heat a container containing sterile milk protein hydrolysate agar to about 50° C. and, using aseptic techniques, dispense approximately 20 ml of the agar into each of the petri dishes containing portions of the specimen. Mix the contents of the petri dishes by rotating the dishes not less than 25 times maintaining them in a horizontal position by sliding them on a flat surface and being careful not to spill the agar over the lips of the dishes. After mixing the contents of the dishes, allow the agar to harden, cover and invert the plates, and incubate them at 35° ± 1° C. for not less than 24 hours. At the end of the incubation period, count the plates containing between 30 and 300 colonies per plate by means of a colony counter. (If neither of the pairs of plates comes within this range, calculate the dilution required and replate.) Calculate the number of bacteria per gram.

4.6 Acceptance/rejection criteria. If any specimen fails to meet any test requirement of this specification, the lot represented shall be rejected.

5. PREPARATION FOR DELIVERY

5.1 Packing, level C. The casein hydrolysate, in the quantities specified in the contract or order, shall be packed in a manner to insure carrier acceptance and safe delivery at destination. Containers shall be in accordance with Uniform Freight Classification Rules, or regulations of other carriers applicable to the mode of transportation.

5.2 Marking. In addition to any marking required by the contract or order, shipments shall be marked in accordance with Standard MIL–STD–129.

6. NOTES

6.1 Intended use. Casein hydrolysates are intended for use as components in the preparation of biological media.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number and date of this specification.

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(b) Grade of casein hydrolysate required.

(c) Net weight of packing desired.

6.3 Nutritional value. Due to unknown factors affecting the growth of various microorganisms and the possible danger in handling these organisms, the Government will conduct all tests on samples to establish the nutritional value of the product. It is estimated that within 30 days after receipt of the pre-award samples or acceptance samples the prospective bidders or suppliers will be notified of the acceptance or the rejection of the product on the basis of nutritional value.

6.4 Batch. A batch is defined as that quantity of material which has been manufactured by some unit chemical process or subjected to some physical mixing operation intended to make the final product substantially uniform.

6.5 Milk protein hydrolysate agar. The milk protein hydrolysate agar, in dehydrated form, may be procured from Baltimore Biological Laboratory Inc., Baltimore, Maryland.

6.6 Copper. A flame photometer may be used for the copper determination.

6.7 Pancreatic casein hydrolysate. The pancreatic casein hydrolysates (grade A and grade B) described in this purchase description are essentially similar to the following products of the Sheffield Chemical Division of National Dairy Products Corporation, Norwich, New York:

N–Z–Amine Type A

N–Z–Amine Type B

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Custodians:
Army—MU
Navy—BUMED
Air Force—AF CSG–33.1

Preparing activity:
Army—MU

Review interest:
Army—MD
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Navy—WP, CG

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